

VIRINCHI SAI ATHMAKURI

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Professional Summary

Applied AI & Machine Learning Engineer specializing in secure AI, enterprise LLM platforms, retrieval-augmented generation, MCP-based tool integration, agentic AI, and cloud-native AI systems. Experience spans AI architecture, secure deployment, retrieval systems, observability, and AI security research across enterprise assistants, SOC copilots, SIEM-style platforms, and multimodal AI applications.

Technical Skills

AI Platform Engineering: Python, FastAPI, Next.js, Streamlit, REST APIs, service design, authentication, audit logging, containerized services

LLM, RAG & Agent Systems: Enterprise RAG, MCP, tool-calling agents, retrieval pipelines, vector search, prompt safeguards, citation-grounded responses, provider abstractions

Secure AI & Security Engineering: JWT, RBAC, secure local deployment, audit trails, prompt-injection defenses, read-only query controls, file access controls, adversarial ML, IDS research

Cloud, Data & Distributed Systems: Docker, Kubernetes, PostgreSQL, PGVector, SQLite, Redpanda/Kafka-style streaming, batch and real-time pipelines, AWS, GCP

MLOps & Observability: Prometheus, Grafana, CI/CD, test automation, MLflow, structured logging, model-serving optimization, deployment verification

Applied ML & Research: Deepfake audio detection, anomaly detection, autoencoders, CNN/LSTM systems, evaluation workflows, intelligent systems prototyping

Professional Experience

Incoming GenAI Engineer

Smarc Technologies LLC

Jul 2026 – Present

- Joining Smarc Technologies LLC in a full-time remote GenAI Engineer internship focused on generative AI solution development, LLM integration, retrieval-augmented generation, AI orchestration, enterprise knowledge pipelines, and scalable AI engineering workflows.
- Scope in the offer letter includes requirements analysis, GenAI application development, foundation model integration, vector database and document indexing pipelines, and MLOps support for deployable AI systems.

Research Engineer

State University of New York Polytechnic Institute

Aug 2023 – Jun 2026

Applied AI & GenAI Systems

Jan 2026 – Jun 2026

- Architected and built **SentinelAI**, an MCP-powered enterprise AI workspace with a FastAPI gateway, Next.js frontend, PGVector-backed retrieval, citation-grounded RAG, JWT/RBAC enforcement, audit logging, Prometheus metrics, Docker, and Kubernetes-ready deployment.
- Engineered secure agent workflows that connect enterprise knowledge, source code, email, calendar, files, and database tools through MCP, with tool traces, approval gates, and provider-agnostic LLM orchestration for enterprise assistant use cases.

Secure Enterprise LLM Platform

Jul 2025 – Dec 2025

- Built a secure enterprise LLM platform for grounded question answering over cybersecurity evidence using FastAPI, Streamlit, LangChain retrieval, FAISS vector storage, JWT authentication, and SQLite-backed audit logging.
- Designed per-user document ingestion, chunking, embedding, and retrieval workflows with citation-backed responses, prompt-injection filtering, checksum-based document handling, and Docker Compose deployment for local or self-hosted operation.

AI Research for Intelligent Systems

Feb 2025 – Jul 2025

- Designed and deployed cloud-native AI pipelines across AWS and GCP for model training, batch processing, and inference services using Docker, Kubernetes, FastAPI, and production-oriented data workflows.
- Integrated enterprise retrieval and GenAI components using AWS Bedrock and Vertex AI to prototype hybrid inference and retrieval architectures for applied research use cases.
- Built autonomous and semi-autonomous AI workflows for planning, retrieval, tool use, and operational task support, emphasizing architecture, modular services, and deployment readiness over isolated experimentation.

Deepfake Audio Detection & AI Security

Aug 2023 – Jul 2024

- Developed deepfake audio detection pipelines and AI security research workflows focused on robust classification, reproducible experimentation, and deployable inference paths for audio forensics use cases.
- Built adversarially tested intrusion-detection models including GACL-DAE, GACL-OCSVM, and GACL-DAE+MLP, validating performance on NSL-KDD and UNSW-NB15 under FGSM and PGD stress testing.
- Contributed secure AI research spanning detection reliability, model robustness, and system-level considerations for using AI in high-trust security environments.

Data & Cloud Engineer

ConnX AI Inc

Apr 2023 – Jul 2023

- Automated observability workflows for UCaaS and SBC production systems using Python and monitoring platforms, improving reliability and accelerating incident isolation.

- Built real-time log analytics and anomaly-detection APIs for continuous ingestion, alerting, and multi-region operational visibility.
- Standardized secure service deployment with containerization and CI/CD automation for production support teams.

Selected AI Systems / Projects

1. SentinelAI — Secure Enterprise AI Workspace

- Flagship system: a secure, MCP-powered enterprise AI workspace built with FastAPI, Next.js, PGVector, RAG, JWT/RBAC, Docker, Kubernetes, Prometheus, and audit logging.

2. SecureAI SOC Copilot

- Built a secure SOC copilot for grounded investigation over security logs and reports using FastAPI, Streamlit, LangChain retrieval, FAISS, JWT authentication, and SQLite audit records.

3. Secure Agentic CloudOps SIEM Platform

- Built a cloud-native SIEM-style platform with FastAPI ingest services, Redpanda streaming, rule-based detection services, and PostgreSQL persistence for events and alerts.

4. PageMint Offline

- Built a privacy-first local AI application that converts documents into Markdown with a standalone user experience, local processing, and optional MCP-based desktop integration.

Education

Master of Science, Network and Computer Security

State University of New York Polytechnic Institute

Bachelor of Engineering, Computer Science

Methodist Engineering College

Certifications

- CompTIA Security+
- AWS Academy Cloud Foundations
- Google Cloud Ready Facilitator
- Coursera Artificial Intelligence & Machine Learning Specialization
- Cisco DevNet, Cybersecurity, and Networking certifications

Selected Publications

- *Video and Audio Deepfake Datasets and Open Issues in Deepfake Technology* — Forensic Sciences, 2024
- *Design and Implementation of an AI Virtual Mouse Using Hand Gesture Recognition* — 2024